Lab task 1

```
struct book
{
    int bookId;
    float price;
    int pages;
};

int main()
{
    int n, choice;
    cout << "Enter number of books: ";
    cin >> n;
    book b[n];
    book *ptr = b;
    cout << "\nENTER BOOKS DETAILS:\n";
    for(int i=0; i<n; i++)
    {
        cout << "\nBook " << i+1 << endl;
        cout << "Enter Book ID: ";
        cin >> (ptr+i)->bookId;
        cout << "Enter Price: ";
        cin >> (ptr+i)->price;
        cout << "Enter Pages: ";
        cin >> (ptr+i)->pages;
    }
    cout << "\nEnter 1 to display most costly book";
    cout << "\nEnter 2 to enter details again: ";
    cin >> choice;
    if(choice == 1)
    {
        float maxprice = ptr->price;

        for(int i=1; i<n; i++)
        {
            if((ptr+i)->price > maxprice)
                maxprice = (ptr+i)->price;
        }

        cout << "\nPrice of most costly book is: " << maxprice << endl;
    }
    else if(choice == 2)
    {
        book newbook;
```

```
C:\Users\Taha's Lappy\OneDrive\Desktop\oop lab 5\lab 5labtask 1.exe
Enter number of books: 2

ENTER BOOKS DETAILS:

Book 1
Enter Book ID: 22
Enter Price: 345
Enter Pages: 55

Book 2
Enter Book ID: 3456
Enter Price: 666
Enter Pages: 55

Enter 1 to display most costly book
Enter 2 to enter details again: 2

Enter Book details again:
Enter Book ID: 234
Enter Price: 123
Enter Pages: 34

You entered:
Book ID: 234
Price: 123
Pages: 34

--------------------------------
Process exited after 20.09 seconds with return value 0
Press any key to continue . . .
```

Lab task 2

labtask1.cpp

```
#include<iostream>
using namespace std;
struct citizen
{
    int CNICID;
    char name[30];
    char city[20];
    int age;
};
int main()
{
    citizen c;
    citizen*ptr;
    ptr=&c;
    cout << "ENTER CNIC ID: ";
    cin >> ptr->CNICID;
    cout << "ENTER  NAME : ";
    cin >> ptr->name;
    cout << " ENTER CITY NAME : ";
    cin >> ptr->city;
    cout << "ENTER AGE : ";
    cin >> ptr->age;
     int choice;
    cout << "\nPress 1 to update city of residence, 0 to display record: ";
    cin >> choice;
    if (choice ==1)
    {
    cout << "Enter new city: ";
    cin >> ptr->city;
    cout << "\nUpdated Citizen Record:\n";
    cout << "CNIC ID: " << ptr->CNICID << endl;
    cout << "Name: " << ptr->name << endl;
    cout << "City: " << ptr->city << endl;
    cout << "Age: " << ptr->age << endl;
   }
    else
    {
    cout << "\nOriginal Citizen Record:\n";
    cout << "CNIC ID: " << ptr->CNICID << endl;
    cout << "Name: " << ptr->name << endl;
    cout << "City: " << ptr->city << endl;
    cout << "Age: " << ptr->age << endl;
    }
  return 0;
```

```
C:\Users\Taha's Lappy\OneDrive\Desktop\oop lab 5\labtask1.exe
ENTER CNIC ID: 46376232
ENTER  NAME : AMNA
 ENTER CITY NAME : LAHORE
ENTER AGE : 19

Press 1 to update city of residence, 0 to display record: 1
Enter new city: KARACHI

Updated Citizen Record:
CNIC ID: 46376232
Name: AMNA
City: KARACHI
Age: 19

--------------------------------
Process exited after 16.68 seconds with return value 0
Press any key to continue . . .
```

Lab task 3

```cpp
#include<iostream>
using namespace std;
struct PetrolPump
{
char name[20];
float pricePerLiter;
int availableLiters;
};
void input (PetrolPump *p)
{
    cout << "ENTER PETROLPUMP NAME : ";
    cin >> p->name;
    cout << "Enter Price per Liter: ";
    cin >> p->pricePerLiter;
    cout << "Enter Available Liters: ";
    cin >> p->availableLiters;
}
void display(PetrolPump *p)
{
    cout << "\n      petrol pump details      " << endl;
    cout << "NAME " << p->name << endl;
    cout << "Price per Liter: " << p->pricePerLiter << endl;
    cout << "Available Liters: " << p->availableLiters << endl;
}
int main()
{
    PetrolPump p;
    input(&p);
    display(&p);
    return 0;
}
```

```
ENTER PETROLPUMP NAME : shell
Enter Price per Liter: 999
Enter Available Liters: 50

      petrol pump details
NAME shell
Price per Liter: 999
Available Liters: 50

--------------------------------
Process exited after 5.585 seconds with return value 0
Press any key to continue . . .
```

Lab task 4

```cpp
#include<iostream>
using namespace std;
struct Bill
 {
    int accountNo;
    char ownerName[30];
    int unitsConsumed;
    float ratePerUnit;
    float totalBill;
};
void inputBill(Bill *b)
{
    cout << "Enter Account Number: ";
    cin >> b->accountNo;
    cout << "Enter Owner Name: ";
    cin >> b->ownerName;
    cout << "Enter Units Consumed: ";
    cin >> b->unitsConsumed;
    cout << "Enter Rate per Unit: ";
    cin >> b->ratePerUnit;
}
void calculateBill(Bill *b)
{
    b->totalBill = b->unitsConsumed * b->ratePerUnit;
    if(b->unitsConsumed > 300)
    {
        b->totalBill = b->totalBill + (b->totalBill * 0.15);
    }
}
void displayBill(Bill *b)
{
    cout << "\n======= ELECTRICITY BILL =======\n";
    cout << "Account No : " << b->accountNo << endl;
    cout << "Owner Name : " << b->ownerName << endl;
    cout << "Units Used : " << b->unitsConsumed << endl;
    cout << "Rate/Unit  : " << b->ratePerUnit << endl;
    cout << "Total Bill : " << b->totalBill << endl;
    cout << "================================\n";
}
int main()
 {
    Bill b;
    cout << "=== LESCO Billing System ===" << endl;
    inputBill(&b);
```

```
C:\Users\Taha's Lappy\OneDrive\Desktop\oop lab 5\LABTASK4.exe
=== LESCO Billing System ===
Enter Account Number: 88378283
Enter Owner Name: SADIA
Enter Units Consumed: 55
Enter Rate per Unit: 333

======= ELECTRICITY BILL =======
Account No : 88378283
Owner Name : SADIA
Units Used : 55
Rate/Unit  : 333
Total Bill : 18315
================================

--------------------------------
Process exited after 12.16 seconds with return value 0
Press any key to continue . . .
```

Hometask 1

```cpp
#include<iostream>
using namespace std;
struct Patient
 {
    int patientID;
    char name[30];
    float temperature;
    int severity;
};
Patient* mostCritical(Patient arr[], int n)
{
    int maxIndex = 0;

    for(int i=1; i<n; i++)
    {
        if(arr[i].severity > arr[maxIndex].severity)
            maxIndex = i;
    }
    cout << "\nMost Critical Patient(s):\n";
    for(int i=0; i<n; i++)
    {
        if(arr[i].severity == arr[maxIndex].severity)
        {
            cout << "ID: " << arr[i].patientID
                 << "  Name: " << arr[i].name
                 << "  Temp: " << arr[i].temperature
                 << "  Severity: " << arr[i].severity << endl;
        }
    }
    return &arr[maxIndex];
}
Patient* findByID(Patient arr[], int n, int id)
{
    for(int i=0; i<n; i++)
    {
        if(arr[i].patientID == id)
            return &arr[i];
    }
    return NULL;
}
void displayPatient(Patient *p)
{
    cout << "ID: " << p->patientID << endl;
    cout << "Name: " << p->name << endl;
    cout << "Temperature: " << p->temperature << endl;
    cout << "Severity: " << p->severity << endl;
}
void inputPatient(Patient *p, int index)
{
    cout << "\nEnter data for Patient " << index+1 << endl;
    cout << "Enter ID: ";
    cin >> p->patientID;
    cout << "Enter Name: ";
    cin >> p->name;
    cout << "Enter Temperature: ";
    cin >> p->temperature;
    cout << "Enter Severity (1-10): ";
```

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C:\Users\Taha's Lappy\OneDrive\Desktop\oop lab 5\HOMETASK1.exe

```
Enter Name: sadia
Enter Temperature: 100
Enter Severity (1-10): 3

Enter data for Patient 3
Enter ID: 333
Enter Name: aima
Enter Temperature: 101
Enter Severity (1-10): 2

Enter data for Patient 4
Enter ID: 444
Enter Name: hooria
Enter Temperature: 12
Enter Severity (1-10): 5

Enter data for Patient 5
Enter ID: 555
Enter Name: hurrain
Enter Temperature: 102
Enter Severity (1-10): 4

Most Critical Patient(s):
ID: 444  Name: hooria  Temp: 12  Severity: 5

Enter Patient ID to update:
```